Quarto & R - Page Management

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Page Management/ Article Layout

Rendering

You can render the document to all formats, or to a specific format using the following commands: - quarto render report.qmd –to all - quarto render report.qmd –to html - quarto render report.qmd –to pdf

https://quarto.org/docs/authoring/article-layout.html

Body Column

Normal Body

Normal-width text here…

if you'd like, you can extend content slightly outside the bounds of the body by creating a div with the .column-body-outset class. For example:

Body Outset

Wider text here…

Body Outset-2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua…

Body Outset-3

magna consectetur eiusmod et dolor dolor do eiusmod elit incididunt lorem labore adipiscing magna sit adipiscing aliqua lorem ipsum sed aliqua aliqua tempor do incididunt tempor magna tempor amet et ipsum ipsum ut elit adipiscing et dolor tempor ipsum tempor consectetur lorem lorem eiusmod eiusmod magna adipiscing dolor consectetur dolor elit elit dolore incididunt elit elit eiusmod consectetur ut magna et incididunt sit labore magna adipiscing eiusmod elit amet lorem aliqua lorem labore incididunt ut amet sed do adipiscing dolor sit aliqua aliqua consectetur labore dolor labore do do dolor sit aliqua incididunt eiusmod elit elit ut adipiscing et sit

Page

If you need even more space for your content, you can use the .column-page class to make the content much wider, though stopping short of extending across the whole document.



Figure 1: Elephant

For computational output, you can specify the page column in your code cell options. For example:

Screen Column

You can have content span the full width of the page with no margin (full bleed). For this, use the .column-screen class or specify column: screen on a code cell. For example:



Figure 2: A full screen image

The following code displays a Leaflet map across the whole page.

```
library(leaflet)
leaflet() %>%  addTiles() %>%  # Add default OpenStreetMap map tiles
  addMarkers(lng=174.768, lat=-36.852, popup="The birthplace of R")
```

Screen Inset

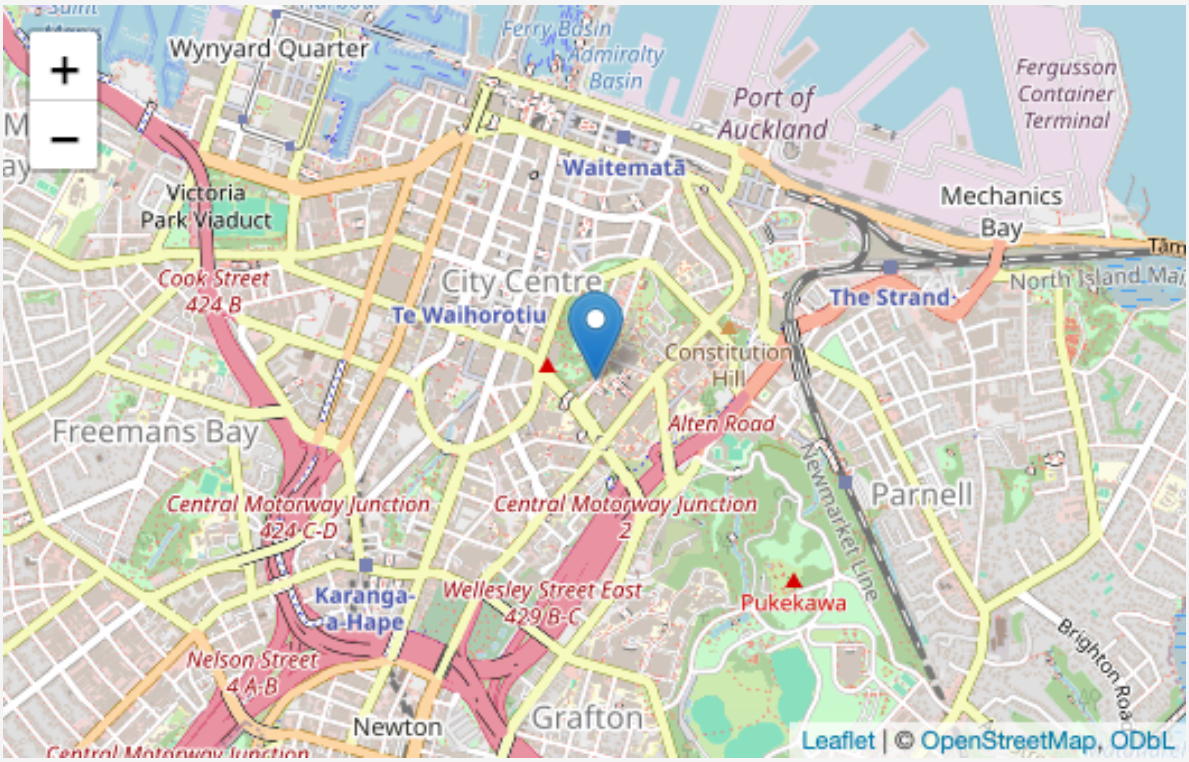
If you'd like a full width appearance, but would like to retain a margin, you can use inset or inset-shaded modifiers on the column. For example:



Figure 3: A full screen image

The inset-shaded modifier results in a block spanning the full width but wrapped with a lightly shaded background. For example:

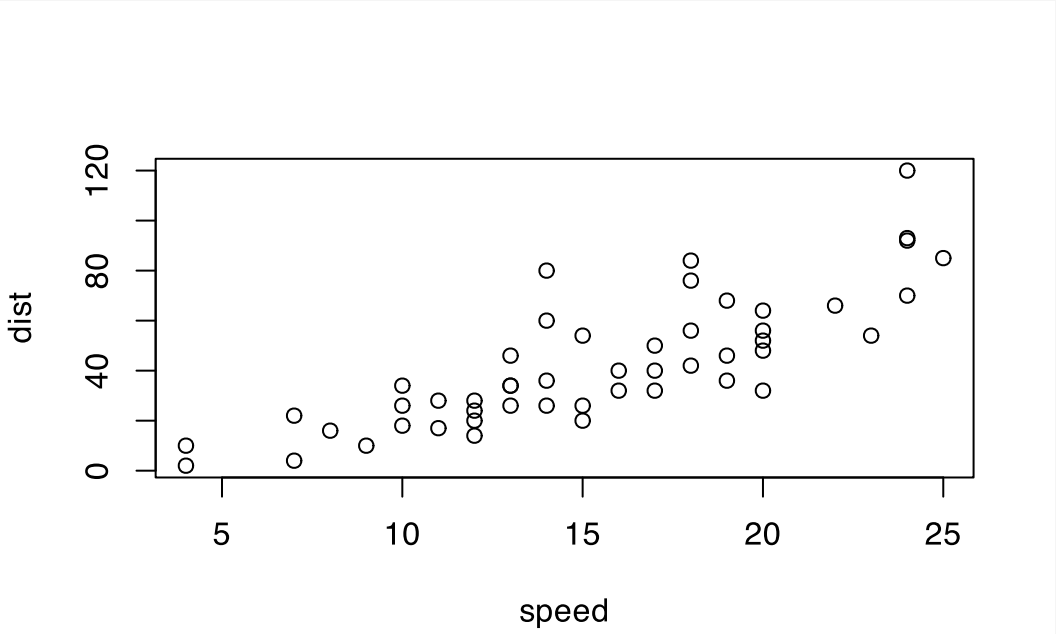
```
library(leaflet)
leaflet() %>%  addTiles() %>%   addMarkers(lng=174.768,
lat=-36.852, popup="The birthplace of R")
```



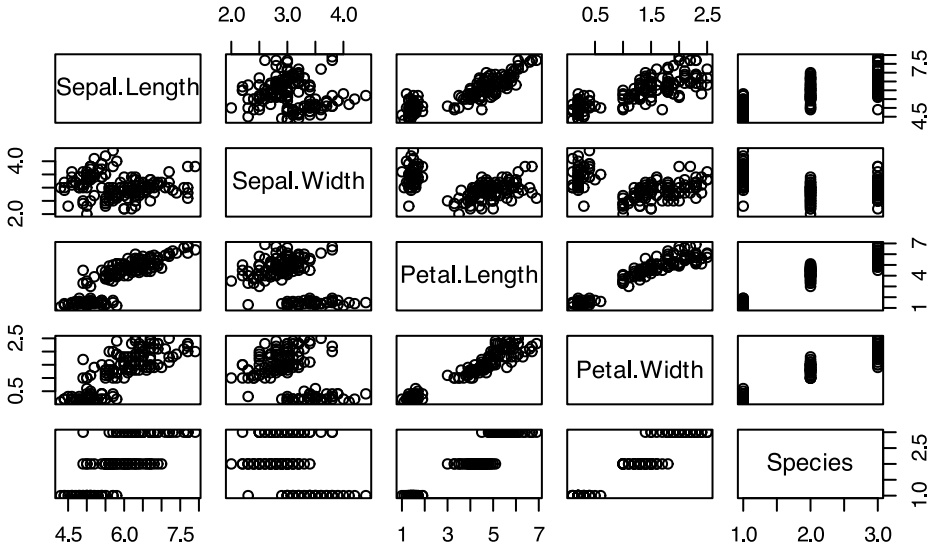

```
#28.5439° N, 77.3333° E - Amity
```

Column layouts like screen-inset-shaded will work with advanced figure layout. For example, it is straightforward to create a multi column presentation of figures that spans the document:

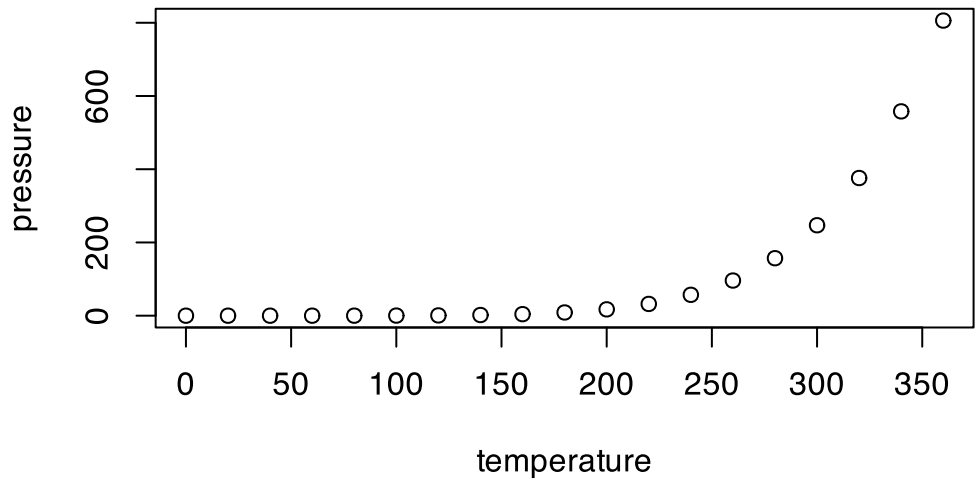
```
plot(cars)
```




```
plot(iris)
```




```
plot(pressure)
```



Margin Content

You can place content within the right margin of Quarto document. For example, here we use the .column-margin class to place an image in the margin:



Figure 4: A margin image

This also works for text content:

We know from *the first fundamental theorem of calculus* that for x in $[a, b]$:

$$\frac{d}{dx}\left(\int_a^x f(u)\,du\right) = f(x).$$

Margin Figures

Figures that you create using code cells can be placed in the margin by using the column: margin code cell option. If the code produces more than one figure, each of the figures will be placed in the margin.

```
library(ggplot2)
mtcars2 <- mtcars
mtcars2$am <- factor(  mtcars$am, labels = c('automatic',
'manual'))
ggplot(mtcars2, aes(hp, mpg, color = am)) +  geom_point()
+  geom_smooth(formula = y ~ x, method = "loess") +
  theme(legend.position = 'bottom')
hist(mtcars2$mpg, main = "Histogram of MPG", xlab = "MPG")
```

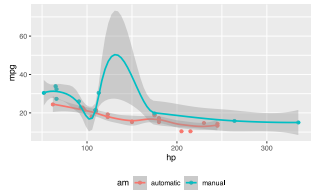


Figure 5: MPG vs horsepower, colored by transmission.

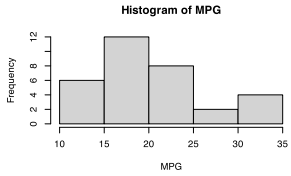


Figure 6: MPG vs horsepower, colored by transmission.

Margin Tables

You can also place tables in the margin of your document by specifying column: margin.

```
knitr::kable(  mtcars[1:6, 1:10], caption = "A table of
the first 6 rows of the mtcars dataset")
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3

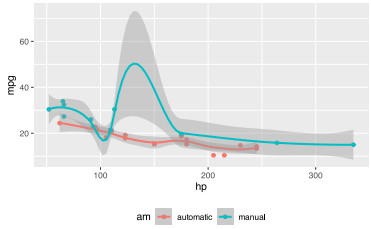
Multiple Outputs

You can also target specific output types (for example, figures) to be placed in the margin. For example, the following code will render a table showing part of the mtcars dataset and produce a plot in the margin next to the table.

```
mtcars2 <- mtcars
mtcars2$am <- factor(  mtcars$am, labels = c('automatic',
'manual'))

knitr::kable(  mtcars[1:6, 1:6])
library(ggplot2)
ggplot(mtcars2, aes(hp, mpg, color = am)) +  geom_point()
+   geom_smooth(formula = y ~ x, method = "loess") +
theme(legend.position = 'bottom')
```

	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21.0	6	160	110	3.90	2.620
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875
Datsun 710	22.8	4	108	93	3.85	2.320
Hornet 4 Drive	21.4	6	258	110	3.08	3.215
Hornet Sportabout	18.7	8	360	175	3.15	3.440
Valiant	18.1	6	225	105	2.76	3.460



Page Breaks

The pagebreak shortcode enables you to insert a native pagebreak into a document (e.g. in LaTeX this would be a , in MS Word a docx-native pagebreak, in HTML a page-break-after: always CSS directive, etc.): No page break when format is html - with HTML - div style="break-before: page;"> page 1 /newpage

page 2

Native pagebreaks are supported for HTML, LaTeX, Context, MS Word, Open Document, ePub, and Typst (for other formats a form-feed character s inserted).

Margin References

Footnotes and the bibliography typically appear at the end of the document, but you can choose to have them placed in the margin by setting the following option1 in the document front matter:

1 You can also position references in other location (such as the bottom of the block, section, or document).

With these options set, footnotes and citations will (respectively) be automatically be placed in the margin of the document rather than the bottom of the page. As an example, when I cite Xie et al. (2018), the citation bibliography entry itself will now appear in the margin.

Asides

Asides allow you to place content aside from the content it is placed in. Asides look like footnotes, but do not include the footnote mark (the superscript number).

This is a span that has the class aside which places it in the margin without a footnote number.

This is a span that has the class `aside` which places it in the margin without a footnote number.

Margin Captions

For figures and tables, you may leave the content in the body of the document while placing the caption in the margin of the document. Using cap-location: margin in a code cell or document front matter to control this. For example:

```
library(ggplot2)
mtcars2 <- mtcars
mtcars2$am <- factor(  mtcars$am, labels = c('automatic',
'manual'))
ggplot(mtcars2, aes(hp, mpg, color = am)) +  geom_point()
+   geom_smooth(formula  =  y  ~  x,  method  =  "loess")  +
theme(legend.position = 'bottom')
```

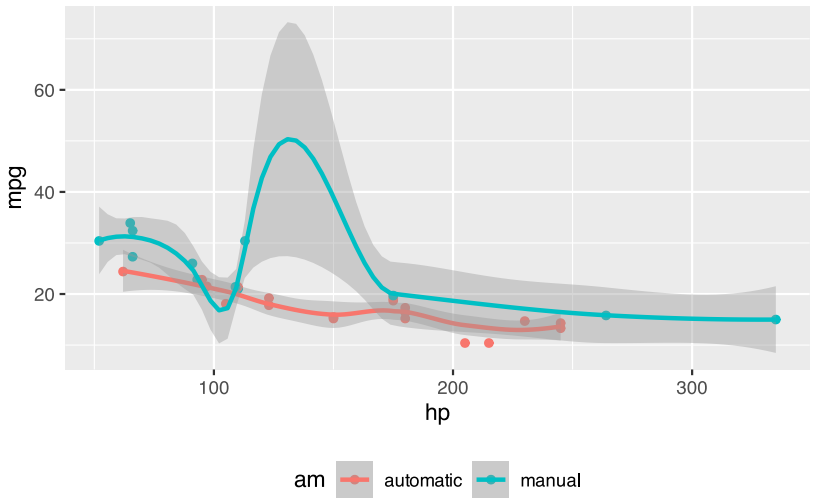


Figure 7: MPG vs horsepower, colored by transmission.

Overflowing Content

You can also extend content outside the body region on only the left or right side of the document by using the right and left versions of the body, page, and screen columns to layout your content. The left or right version of these columns are as follows:

.column-body-outset-right

.column-page-inset-right

.column-page-right

.column-screen-inset-right

.column-screen-right

.column-body-outset-left

.column-page-inset-left

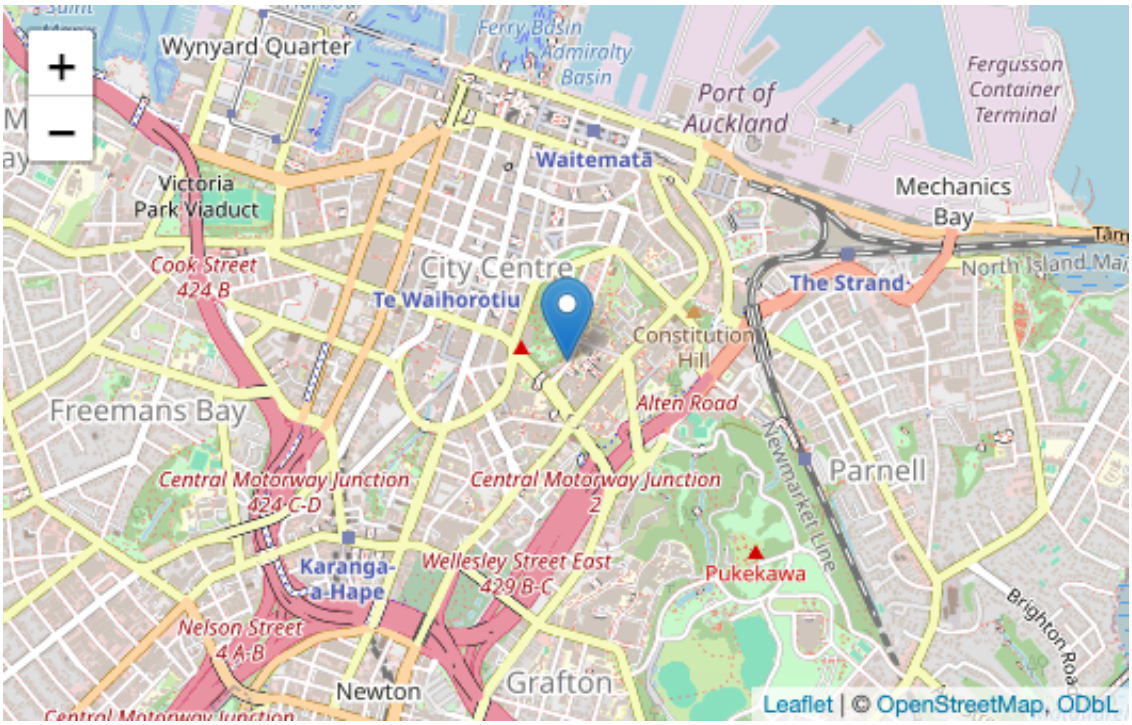
.column-page-left

.column-screen-inset-left

.column-screen-left

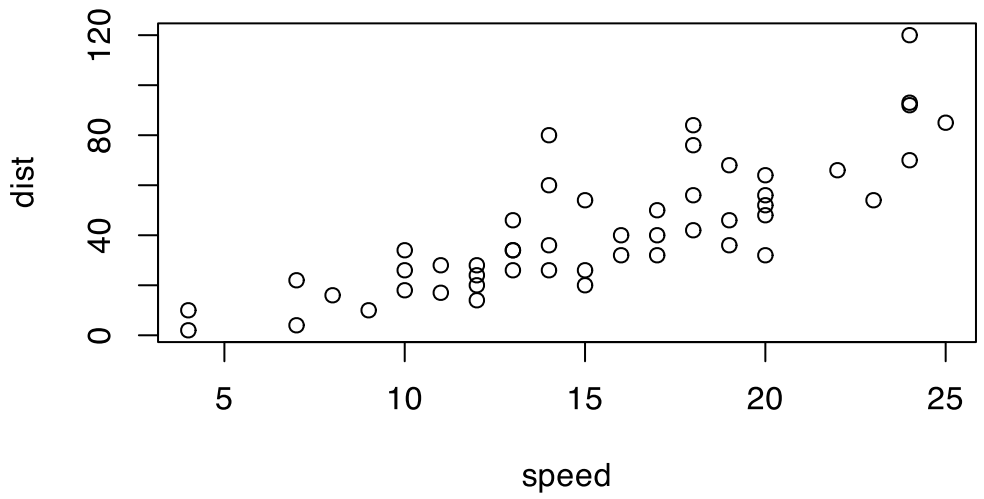
Use a div with one of the above classes to align content into one of the overflow regions. This also works using the column option of executable code cells:

```
library(leaflet)
leaflet() %>%
  addTiles() %>%  # Add default OpenStreetMap map tiles
     addMarkers(lng=174.768,   lat=-36.852,   popup="The
birthplace of R")
```



Code Cell Options You can also set layout column on specific code cells. This controls the layout of content that is produced by the code cell.

```
plot(cars)
```



This will appear in portrait mode.

This will appear in landscape.

This will appear in portrait mode again.